

Core JS + Node Assignment

1. Variables and Console

Write a Node.js program to:

- Store your name, age, and city.
- Display them using `console.log()`.

2. Loop Program

Display numbers from:

1 to 50

using a for loop, foreach, while loop.

3. Array Operations

Create an array of student names.

Perform:

- Add element
- Remove element
- Display all elements

4. OS Module

Display:

- Hostname
- Platform
- CPU Architecture
- Total Memory

5. File System Module

Create a file:

student.txt

Write student information and read it back.

6. NPM Practice

Install:

Nodemon similar how you run your application using node (node file.js)

Explain its use and run a Node.js application using it.

7. MongoDB CRUD

Perform MongoDB CRUD operation for Student Domain

Follow the schema below:

Field Name	Data Type	Description
id	Number	Unique Student ID
name	String	Student Name
email	String	Student Email
course	String	Course Enrolled
age	Number	Student Age
city	String	Student City
fees	Number	Course Fees
isActive	Boolean	Student Status

Operations:

1. Add a New Student
2. View All Students
3. Delete Multiple Student
4. Update Student Details

Viva / Interview-Oriented Questions

1. Difference between Node.js and JavaScript?
2. What is Event Loop?
3. What is Callback Hell?
4. Difference between Promise and Async/Await?

5. What is Middleware in Express?
6. Difference between MongoDB and MySQL?
7. What is Mongoose?
8. What is REST API?
9. Difference between require() and import?
10. What is package.json?
11. What is npm install?
12. What is the purpose of .env?
13. What is the difference between Authentication and Authorization?